

ON THE DISTRIBUTION OF THE LIKELIHOOD RATIO TEST STATISTIC FOR COMPOUND SYMMETRY

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SUMMARY

The problem of testing the hypothesis on the compound symmetry of the covariance matrix of the multivariate normal distribution plays a useful role in areas like medical research and psychometrics. Votaw (1948) derived the likelihood ratio statistic for testing the above hypothesis and also obtained the moments of this statistic. It is very complicated to compute the percentage points of the distribution of the above statistic using exact expressions except in very special cases. So, the authors approximated this distribution with a suitable distribution belonging to the family of Pearson type distributions. It is found that the accuracy of this approximation is sufficient for practical purposes. Using the above approximation, percentage points of the distribution of the test statistic for compound symmetry are computed.

1. Statement of the problem and preliminaries

Let $\underline{X}' = (\underline{X}'_1, \dots, \underline{X}'_k)$ be distributed as a t-variate normal with mean vector $\underline{\mu}$ and covariance matrix Σ . Also, let $\underline{X}_i$ be of order 1×1 for $i \leq b$ and $p_1 \times 1, p_2 \times 1$ respectively for $i = b + 1, b + 2$. In addition, $t = b + \sum_{i=1}^2 p_i$, $k = b + 2$, and

$$E \{ (\underline{X}_i - \underline{\mu}_i) (\underline{X}_j' - \underline{\mu}_j') \} = \Sigma_{ij} = (\sigma_{ijuv}).$$

In this paper, we study the distribution of the likelihood ratio statistic for testing H where H denotes the hypothesis that the variances are equal and the covariances are equal within any set of variables, and the covariances of any two variables belonging to any two distinct sets are equal. Throughout this paper, we assume that $p_i \geq 2$ for $i = 1, 2$.

2. Distribution of the Likelihood Ratio Test Statistic

Let $\underline{X}_{i\alpha}$ be the α -th independent observation on $\underline{X}_i$ for $i = 1, \dots, k$ and $\alpha = 1, 2, \dots, N$.

Also, let $\nu_{ij} = \sum_{\alpha=1}^N (X_{i\alpha} - \bar{X}_{i\cdot})(X_{j\alpha} - \bar{X}_{j\cdot})$ where $\bar{X}_{i\cdot} = N^{-1} \sum_{\alpha=1}^N X_{i\alpha}$. The likelihood

ratio test statistic for H is known to be

$$(2.1) \quad L = \frac{|V|}{|\tilde{V}|}$$

where $V = (\nu_{ij})$, $\tilde{V} = (\tilde{\nu}_{ij})$, $\tilde{\nu}_{ss'} = \nu_{ss'}$, $\tilde{\nu}_{si_a} = \frac{1}{p_a} \sum_{j_a} \nu_{sj_a}$,

$$\tilde{\nu}_{i_a i_a} = \frac{1}{p_a} \sum_{j_a} \nu_{j_a j_a}, \quad \tilde{\nu}_{i_a j_a} = \frac{1}{p_a(p_a - 1)} \sum_{\ell_a \neq m_a} \nu_{\ell_a m_a},$$

$$\tilde{\nu}_{i_a j_{a'}} = \frac{1}{p_a p_{a'}} \sum_{\ell_a m_{a'}} \nu_{\ell_a m_{a'}},$$

$$s, s' = 1, \dots, b; a, a' = 1, 2; a \neq a'; i_a, \ell_a, j_a, m_a$$

$$= b + \bar{p}_a + 1, \dots, b + \bar{p}_{a+1};$$

$$\bar{p}_a = p_1 + \dots + p_{a-1}; \bar{p}_1 = 0.$$

The d-th moment of L is given by

$$(2.2) \quad E(L^d) = \prod_{\alpha=1}^2 \frac{p_{\alpha} - 1}{s_{\alpha} - 1} \left\{ \frac{\left(\frac{N - k - s_{\alpha} - \bar{p}_{\alpha} + \alpha - 1}{2} \right)_d}{\left(\frac{N-1}{2} + \frac{s_{\alpha}-1}{p_{\alpha}-1} \right)_d} \right\}$$

where $(T)_d = \Gamma(T+d)/\Gamma(T)$, $\Gamma(T)$ is the complete gamma function. The statistic L and its moments were derived by Votaw (1948). According to the likelihood ratio test we accept or reject H according as $L \geq c$ where

$$(2.3) \quad P[L \geq c|H] = (1 - \alpha).$$

Using the first four moments, the distribution of $L^{1/g}$ is approximated with a suitable Pearson type distribution where g is a properly chosen integer. It is found that Pearson's Type I curve gives the best fit, among the family of Pearson type distributions. The

density of Pearson's Type I distribution is given by

$$(2.4) \quad f(x) = \frac{1}{\beta(\alpha+1, \epsilon+1)(\sigma-\sigma_0)^{\alpha+\epsilon+1}} (x-\sigma_0)^\alpha (\sigma-x)^\epsilon, \quad \sigma_0 \leq x \leq \sigma.$$

Box (1949) derived asymptotic expressions for the distributions of a class of likelihood ratio test statistics in multivariate analysis. The test statistic L given by (2.1) belongs to this class. In Table I, we give a comparison of values obtained by Pearson type approximation with the corresponding values obtained by Box's asymptotic expression of order N^{-13} . In the appendix we give the coefficients of this expression up to order N^{-13} since Box gave the first few terms only.

If we consider the d -th moment of $L^{N/2}$, we have the following in the notations of the appendix:

$$a = b = p_1 + p_2 - 2, Y_i = X_i = \frac{N}{2}, \xi_i = -\frac{1}{2}(b + 2 + i)$$

for $i = 1, 2, \dots, p_1 + p_2 - 2$ and

$$\eta_i = \begin{cases} \frac{i-1}{p_1-1} - \frac{1}{2} & \text{for } i = 1, 2, \dots, p_1-1 \\ \frac{i-p_1}{p_2-1} - \frac{1}{2} & \text{for } i = p_1, p_1 + 1, \dots, p_1 + p_2 - 2 \end{cases}$$

In Table I, α_1 is the value of α if we use Pearson type approximation whereas α_2 is the value of α if we use the asymptotic expression of order N^{-13} . Also, c and α are defined by Equation (2.3).

When $b = 0$, Consul (1969) and Mathai and Rathie (1970) gave exact expressions for the distribution of L but these expressions are complicated to compute. Danford, Rafferty and Votaw (1950) computed the exact percentage points for $b = 1(1)4$, $p_1 = 2, 3$; $b = 1(1)4$, $p_1 = 2$, $p_2 = 2$; $b = 0$, $p_1 = 2$, $p_2 = 3$. In Table II the entries under the column "Approx" are the percentage points obtained by us using Pearson type approximations whereas the values under the column "Exact" are the exact values obtained by Danford, Rafferty and Votaw (1950).

Tables I and II indicate that the accuracy of Pearson type approximation is sufficient for practical purposes. So, using Pearson type approximation, we computed the values of c for $\alpha = 0.05$, $(p_1, p_2) = (2, 2), (2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (4, 4)$, $b = 0, 1, 2, 3$, $M = 1(1) 20(2) 30(5) 45, 50, 60$, and $M = N - p_1 - p_2 - b - 4$. These values are given in Table III.

TABLE I
COMPARISON OF PEARSON TYPE APPROXIMATION
WITH ASYMPTOTIC EXPANSION OF ORDER $N^{-1/3}$

$b=2, p_1=2, p_2=4$				$b=0, p_1=3, p_2=4$		
N	c	α_1	α_2	c	α_1	α_2
20	0.0902	0.05	0.0495	0.1096	0.05	0.0495
20	0.0582	0.01	0.0099	0.0728	0.01	0.0098
30	0.2365	0.05	0.0499	0.2584	0.05	0.0498
30	0.1821	0.01	0.0100	0.2014	0.01	0.0100
40	0.3564	0.05	0.0499	0.3767	0.05	0.0499
40	0.2958	0.01	0.0100	0.3147	0.01	0.0100
60	0.5179	0.05	0.0500	0.5340	0.05	0.0500
60	0.4599	0.01	0.0100	0.4758	0.01	0.0100

TABLE II
COMPARISON OF PEARSON TYPE APPROXIMATION
WITH EXACT VALUES FOR THE PERCENTAGE POINTS OF L

		$b=0, p_1=2, p_2=3$		$b=2, p_1=2, p_2=2$	
N	α	Approx	Exact	Approx	Exact
10	0.05	0.0569	0.057	0.0430	0.043
10	0.01	0.0259	0.026	0.0170	0.017
15	0.05	0.2030	0.20	0.2009	0.20
15	0.01	0.1320	0.13	0.1273	0.13
20	0.05	0.3301	0.33	0.3381	0.34
20	0.01	0.2451	0.24	0.2490	0.25

TABLE III
PERCENTAGE POINTS OF THE DISTRIBUTION OF L ($\alpha = 0.05$)

b=0

(p_1, p_2)	(2,2)	(2,3)	(2,4)	(2,5)	(3,3)	(3,4)	(4,4)
M							
1	.1397	.0569	.0240	.0102	.0239	.0102	.0044
2	.1886	.0847	.0388	.0178	.0388	.0178	.0081
3	.2350	.1142	.0561	.0274	.0560	.0273	.0132
4	.2782	.1443	.0751	.0386	.0750	.0386	.0196
5	.3178	.1741	.0951	.0513	.0950	.0512	.0271
6	.3540	.2030	.1158	.0649	.1157	.0649	.0357
7	.3871	.2310	.1366	.0794	.1365	.0793	.0452
8	.4173	.2577	.1574	.0944	.1573	.0943	.0554
9	.4449	.2831	.1780	.1097	.1779	.1096	.0662
10	.4702	.3072	.1981	.1252	.1980	.1251	.0775
11	.4935	.3301	.2178	.1408	.2177	.1407	.0890
12	.5148	.3518	.2369	.1563	.2368	.1562	.1009
13	.5346	.3723	.2554	.1717	.2553	.1716	.1129
14	.5528	.3917	.2733	.1868	.2732	.1868	.1250
15	.5698	.4100	.2906	.2018	.2905	.2017	.1371
16	.5855	.4274	.3072	.2164	.3071	.2164	.1492
17	.6002	.4439	.3233	.2308	.3232	.2307	.1613
18	.6138	.4595	.3387	.2448	.3386	.2447	.1733
19	.6265	.4743	.3536	.2584	.3535	.2584	.1851
20	.6385	.4884	.3679	.2718	.3678	.2718	.1968
22	.6604	.5145	.3949	.2975	.3949	.2974	.2197
24	.6799	.5383	.4200	.3217	.4199	.3217	.2419
26	.6972	.5599	.4432	.3447	.4431	.3446	.2633
28	.7126	.5977	.4648	.3664	.4647	.3663	.2838
30	.7267	.5799	.4848	.3868	.4847	.3868	.3034
35	.7565	.6368	.5293	.4332	.5292	.4331	.3490
40	.7805	.6691	.5669	.4735	.5669	.4735	.3898
45	.8002	.6962	.5992	.5088	.5992	.5088	.4262
50	.8168	.7194	.6271	.5399	.6270	.5399	.4589
60	.8428	.7564	.6728	.5920	.6728	.5919	.5149

TABLE III (Continued)

 $b = 1$

(p_1, p_2)	(2,2)	(2,3)	(2,4)	(2,5)	(3,3)	(3,4)	(4,4)
M							
1	.0973	.0372	.0149	.0061	.0149	.0061	.0025
2	.1365	.0577	.0251	.0110	.0251	.0110	.0048
3	.1755	.0804	.0375	.0175	.0375	.0175	.0081
4	.2130	.1044	.0517	.0255	.0516	.0255	.0124
5	.2487	.1290	.0671	.0347	.0671	.0347	.0177
6	.2822	.1536	.0835	.0450	.0835	.0449	.0239
7	.3135	.1779	.1005	.0561	.1005	.0561	.0309
8	.3427	.2016	.1178	.0680	.1178	.0679	.0385
9	.3697	.2246	.1353	.0803	.1352	.0803	.0468
10	.3950	.2468	.1527	.0930	.1527	.0930	.0557
11	.4185	.2682	.1700	.1060	.1699	.1060	.0649
12	.4404	.2886	.1870	.1192	.1870	.1191	.0745
13	.4608	.3083	.2038	.1324	.2037	.1323	.0844
14	.4799	.3270	.2201	.1456	.2201	.1456	.0945
15	.4977	.3450	.2361	.1588	.2361	.1587	.1048
16	.5145	.3622	.2517	.1719	.2516	.1718	.1152
17	.5302	.3786	.2668	.1848	.2667	.1847	.1256
18	.5449	.3943	.2815	.1975	.2814	.1975	.1361
19	.5588	.4093	.2958	.2101	.2957	.2100	.1465
20	.5719	.4236	.3096	.2224	.3095	.2224	.1569
22	.5959	.4504	.3360	.2464	.3359	.2463	.1775
24	.6175	.4750	.3607	.2694	.3607	.2694	.1978
26	.6368	.4977	.3840	.2914	.3839	.2914	.2175
28	.6545	.5185	.4057	.3125	.4056	.3124	.2367
30	.6705	.5378	.4261	.3325	.4261	.3324	.2554
35	.7046	.5801	.4719	.3785	.4719	.3785	.2992
40	.7325	.6155	.5114	.4194	.5114	.4193	.3391
45	.7555	.6455	.5456	.4556	.5456	.4555	.3755
50	.7750	.6712	.5755	.4878	.5755	.4878	.4084
60	.8059	.7129	.6252	.5425	.6251	.5425	.4657

TABLE III (Continued)

$b = 2$

(p_1, p_2)	(2,2)	(2,3)	(2,4)	(2,5)	(3,3)	(3,4)	(4,4)
M							
1	.0720	.0258	.0098	.0038	.0098	.0038	.0015
2	.1039	.0412	.0170	.0071	.0170	.0071	.0030
3	.1366	.0590	.0261	.0117	.0261	.0117	.0052
4	.1692	.0783	.0369	.0174	.0368	.0174	.0082
5	.2009	.0986	.0489	.0242	.0489	.0242	.0119
6	.2314	.1194	.0620	.0320	.0620	.0320	.0164
7	.2603	.1404	.0758	.0407	.0758	.0407	.0216
8	.2878	.1612	.0902	.0501	.0902	.0500	.0274
9	.3136	.1818	.1050	.0600	.1050	.0600	.0339
10	.3381	.2019	.1200	.0705	.1200	.0704	.0408
11	.3611	.2215	.1351	.0813	.1350	.0813	.0482
12	.3827	.2406	.1501	.0924	.1501	.0923	.0560
13	.4030	.2590	.1651	.1037	.1650	.1037	.0641
14	.4223	.2768	.1798	.1151	.1798	.1151	.0726
15	.4404	.2940	.1944	.1266	.1943	.1266	.0812
16	.4575	.3105	.2087	.1382	.2087	.1382	.0900
17	.4736	.3265	.2227	.1497	.2227	.1497	.0990
18	.4890	.3418	.2365	.1612	.2364	.1611	.1081
19	.5034	.3566	.2499	.1725	.2499	.1725	.1172
20	.5170	.3708	.2630	.1838	.2630	.1838	.1264
22	.5425	.3977	.2883	.2060	.2883	.2059	.1448
24	.5653	.4225	.3123	.2275	.3122	.2274	.1631
26	.5860	.4456	.3350	.2483	.3350	.2482	.1812
28	.6050	.4671	.3565	.2683	.3564	.2682	.1989
30	.6223	.4869	.3767	.2876	.3768	.2875	.2163
35	.6596	.5310	.4229	.3325	.4229	.3325	.2578
40	.6905	.5684	.4633	.3730	.4634	.3730	.2964
45	.7162	.6004	.4987	.4094	.4987	.4093	.3319
50	.7379	.6283	.5300	.4421	.5298	.4422	.3647
60	.7729	.6736	.5824	.4985	.5824	.4985	.4223

TABLE III (Continued)

b = 3

(P ₁ , P ₂)	(2,2)	(2,3)	(2,4)	(2,5)	(3,3)	(3,4)	(4,4)
M							
1	.0555	.0187	.0067	.0025	.0067	.0025	.0009
2	.0818	.0305	.0119	.0048	.0119	.0048	.0019
3	.1097	.0446	.0188	.0080	.0188	.0080	.0034
4	.1380	.0603	.0270	.0122	.0270	.0122	.0055
5	.1662	.0772	.0365	.0173	.0365	.0173	.0082
6	.1936	.0949	.0470	.0233	.0470	.0233	.0115
7	.2202	.1130	.0584	.0301	.0584	.0301	.0154
8	.2458	.1313	.0704	.0376	.0704	.0376	.0199
9	.2702	.1496	.0829	.0457	.0829	.0457	.0249
10	.2934	.1677	.0958	.0543	.0957	.0542	.0304
11	.3155	.1855	.1088	.0633	.1088	.0633	.0364
12	.3365	.2031	.1221	.0726	.1220	.0726	.0427
13	.3564	.2202	.1353	.0823	.1353	.0823	.0494
14	.3754	.2368	.1486	.0921	.1485	.0921	.0564
15	.3933	.2531	.1618	.1022	.1617	.1021	.0637
16	.4104	.2688	.1748	.1123	.1747	.1123	.0712
17	.4267	.2841	.1877	.1225	.1877	.1225	.0788
18	.4421	.2989	.2005	.1328	.2004	.1327	.0867
19	.4568	.3132	.2129	.1430	.2130	.1430	.0946
20	.4708	.3271	.2253	.1532	.2252	.1532	.1027
22	.4967	.3534	.2492	.1735	.2492	.1734	.1190
24	.5205	.3781	.2721	.1933	.2720	.1933	.1354
26	.5422	.4011	.2941	.2128	.2940	.2128	.1518
28	.5621	.4228	.3150	.2317	.3149	.2316	.1681
30	.5802	.4429	.3349	.2501	.3349	.2500	.1842
35	.6200	.4882	.3808	.2934	.2806	.2934	.2232
40	.6529	.5268	.4213	.3330	.4212	.3330	.2601
45	.6807	.5604	.4573	.3692	.4573	.3691	.2945
50	.7046	.5896	.4894	.4020	.4892	.4019	.3266
60	.7429	.6381	.5438	.4592	.5438	.4592	.3839

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REFERENCES

- ABRAMOWITZ, M. and STEGUN, I.A. (1970). *Handbook of Mathematical Functions*. (Ninth Printing). U.S. Government Printing Office. Washington, D.C.
- BOX, G.E.P. (1949). A General Distribution Theory for a Class of Likelihood Criteria. *Biometrika*, **36**, 317-346.
- CONSUL, P.C. (1969). The Exact Distributions of Likelihood Criteria for Different Hypotheses. In *Multivariate Analysis - II* (P.R. Krishnaiah, editor), 171-181, Academic Press, Inc., New York.
- DANFORD, M.B., RAFFERTY, J.A. and VOTAW, D.F. Jr. (1950). Critical Values of Certain Compound Symmetry Criteria. *U.S.A.F. School of Aviation Medicine, Project 21-49-002, Report No. 4*.
- KENDALL, M.G. and STUART, A. (1947). *The Advanced Theory of Statistics*, (Third Edition). Hafner Publishing Company, New York.
- MATHAI, A.M. and RATHIE, P.N. (1970). The Exact Distribution of Votaw's Criteria. *Ann. Inst. Statist. Math.* **22**, 89-116.
- VOTAW, D.F. Jr. (1948). Testing Compound Symmetry in a Normal Multivariate Distribution. *Ann. Math. Statist.* **19**, 447-473.

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APPENDIX

Let $W(0 \leq W \leq 1)$ be a random variable whose moments are given by

$$E(W^h) = C \left[\prod_{j=1}^b Y_j^{Y_j} / \prod_{i=1}^a X_i^{X_i} \right] \frac{\prod_{i=1}^a \Gamma[X_i(1+h) + \xi_i]}{\prod_{j=1}^b \Gamma[Y_j(1+h) + \eta_j]}$$

where C is chosen such that $E(W^0) = 1$ and $\sum_{i=1}^a X_i = \sum_{j=1}^b Y_j$. Box (1949) gave an asymptotic expression for the distribution of $-2\rho \log W$, where $\rho(0 \leq \rho \leq 1)$ is an arbitrary constant.

In this appendix we give asymptotic expression for the distribution of $-2\rho \log W$ with remaining term of order N^{-13} where ρ is chosen such that $\omega_1 = 0$, ω_r is given by

$$\omega_r = \frac{(-1)^{r+1}}{r(r+1)} \left\{ \sum_{i=1}^a \frac{B_{r+1}(\beta_i + \xi_i)}{(\rho X_i)^r} - \sum_{j=1}^b \frac{B_{r+1}(\epsilon_j + \eta_j)}{(\rho Y_j)^r} \right\}$$

$\beta_j = (1 - \rho)X_j$, $\epsilon_j = (1 - \rho)Y_j$ and $B_r(h)$ is the Bernoulli polynomial of degree r and order unity (see Abrahamowitz and Stegun (1970), p. 809).

Let $g_\nu(z)$ be the probability density function (p.d.f.) of the χ^2 distribution with ν degrees of freedom and N be the sample size. The p.d.f. of $-2\rho \log W$ is given by

$$l(z) = \sum_{r=0}^{12} S_r(z) + O(N^{-13})$$

where $f = -2 \left\{ \sum_{i=1}^a \xi_i - \sum_{j=1}^b \eta_j - \frac{1}{2}(a - b) \right\}$ and where

$$S_0(z) = g_f(z), S_1(z) = 0, S_2(z) = \omega_2[g_{f+4}(z) - g_f(z)],$$

$$S_3(z) = \omega_3[g_{f+6}(z) - g_f(z)],$$

$$S_4(z) = \omega_4[g_{f+8}(z) - g_f(z)] + \frac{1}{2}\omega_2^2[g_{f+8}(z) - 2g_{f+4}(z) + g_f(z)],$$

$$S_5(z) = \omega_5[g_{f+10}(z) - g_f(z)] + \omega_2\omega_3[g_{f+10}(z) - g_{f+6}(z) - g_{f+4}(z) + g_f(z)],$$

$$S_6(z) = \omega_6[g_{f+12}(z) - g_f(z)] + \omega_4\omega_2[g_{f+12}(z) - g_{f+8}(z) - g_{f+4}(z) + g_f(z)]$$

$$+ \omega_3^2[\frac{1}{2}g_{f+12}(z) - g_{f+6}(z) + \frac{1}{2}g_f(z)]$$

$$+ \omega_2^3[\frac{1}{6}g_{f+12}(z) - \frac{1}{2}g_{f+8}(z) + \frac{1}{2}g_{f+4}(z) - \frac{1}{6}g_f(z)],$$

$$S_7(z) = \omega_7[g_{f+14}(z) - g_f(z)] + \omega_5\omega_2[g_{f+14}(z) - g_{f+10}(z) - g_{f+4}(z) + g_f(z)]$$

$$+ \omega_4\omega_3[g_{f+14}(z) - g_{f+8}(z) - g_{f+6}(z) + g_f(z)]$$

$$+ \omega_3\omega_2^2[\frac{1}{2}g_{f+14}(z) - g_{f+10}(z) - \frac{1}{2}g_{f+8}(z) + \frac{1}{2}g_{f+6}(z) + g_{f+4}(z) - \frac{1}{2}g_f(z)],$$

$$S_8(z) = \omega_8[g_{f+16}(z) - g_f(z)] + \omega_6\omega_2[g_{f+16}(z) - g_{f+12}(z) - g_{f+4}(z) + g_f(z)]$$

$$+ \omega_5\omega_3[g_{f+16}(z) - g_{f+10}(z) - g_{f+6}(z) + g_f(z)]$$

$$+ \omega_4^2[\frac{1}{2}g_{f+16}(z) - g_{f+8}(z) + \frac{1}{2}g_f(z)]$$

$$+ \omega_4\omega_2^2[\frac{1}{2}g_{f+16}(z) - g_{f+12}(z) + g_{f+4}(z) - \frac{1}{2}g_f(z)]$$

$$\begin{aligned}
 & + \omega_3^2 \omega_2 \left[\frac{1}{2} g_{f+16}(z) - \frac{1}{2} g_{f+12}(z) - g_{f+10}(z) + g_{f+6}(z) + \frac{1}{2} g_{f+4}(z) - \frac{1}{2} g_f(z) \right] \\
 & + \omega_2^4 \left[\frac{1}{24} g_{f+16}(z) - \frac{1}{6} g_{f+12}(z) + \frac{1}{4} g_{f+8}(z) - \frac{1}{6} g_{f+4}(z) + \frac{1}{24} g_f(z) \right], \\
 S_9(z) = & \omega_9 [g_{f+18}(z) - g_f(z)] + \omega_7 \omega_2 [g_{f+18}(z) - g_{f+14}(z) - g_{f+4}(z) + g_f(z)] \\
 & + \omega_6 \omega_3 [g_{f+18}(z) - g_{f+12}(z) - g_{f+6}(z) + g_f(z)] \\
 & + \omega_5 \omega_4 [g_{f+18}(z) - g_{f+10}(z) - g_{f+8}(z) + g_f(z)] \\
 & + \omega_5 \omega_2^2 \left[\frac{1}{2} g_{f+18}(z) - g_{f+14}(z) + \frac{1}{2} g_{f+10}(z) - \frac{1}{2} g_{f+18}(z) + g_{f+4}(z) - \frac{1}{2} g_f(z) \right] \\
 & + \omega_4 \omega_3 \omega_2 [g_{f+18}(z) - g_{f+14}(z) - g_{f+12}(z) - g_{f+10}(z) + g_{f+8}(z) + g_{f+6}(z) \\
 & \quad + g_{f+4}(z) - g_f(z)] \\
 & + \omega_3^3 \left[\frac{1}{6} g_{f+18}(z) - \frac{1}{2} g_{f+12}(z) + \frac{1}{2} g_{f+6}(z) - \frac{1}{6} g_f(z) \right] \\
 & + \omega_3 \omega_2^3 \left[\frac{1}{6} g_{f+18}(z) - \frac{1}{2} g_{f+14}(z) - \frac{1}{6} g_{f+12}(z) + \frac{1}{2} g_{f+10}(z) \right. \\
 & \quad \left. + \frac{1}{2} g_{f+8}(z) - \frac{1}{6} g_{f+6}(z) - \frac{1}{2} g_{f+4}(z) + \frac{1}{6} g_f(z) \right], \\
 S_{10}(z) = & \omega_{10} [g_{f+20}(z) - g_f(z)] + \omega_8 \omega_2 [g_{f+20}(z) - g_{f+16}(z) - g_{f+4}(z) + g_f(z)] \\
 & + \omega_7 \omega_3 [g_{f+20}(z) - g_{f+14}(z) - g_{f+6}(z) + g_f(z)] \\
 & + \omega_6 \omega_4 [g_{f+20}(z) - g_{f+12}(z) - g_{f+8}(z) + g_f(z)] \\
 & + \omega_6 \omega_2^2 \left[\frac{1}{2} g_{f+20}(z) - g_{f+16}(z) + \frac{1}{2} g_{f+12}(z) - \frac{1}{2} g_{f+8}(z) \right. \\
 & \quad \left. + g_{f+4}(z) - \frac{1}{2} g_f(z) \right] \\
 & + \omega_5^2 \left[\frac{1}{2} g_{f+20}(z) - g_{f+10}(z) + \frac{1}{2} g_f(z) \right] \\
 & + \omega_5 \omega_3 \omega_2 [g_{f+20}(z) - g_{f+16}(z) - g_{f+14}(z) + g_{f+6}(z) + g_{f+4}(z) - g_f(z)] \\
 & + \omega_4^2 \omega_2 \left[\frac{1}{2} g_{f+20}(z) - \frac{1}{2} g_{f+16}(z) - g_{f+12}(z) + g_{f+8}(z) + \frac{1}{2} g_{f+4}(z) - \frac{1}{2} g_f(z) \right] \\
 & + \omega_4 \omega_3^2 \left[\frac{1}{2} g_{f+20}(z) - g_{f+14}(z) - \frac{1}{2} g_{f+12}(z) + \frac{1}{2} g_{f+8}(z) + g_{f+6}(z) - \frac{1}{2} g_f(z) \right]
 \end{aligned}$$

$$\begin{aligned}
& + \omega_3^2 \omega_2^2 \left[\frac{1}{4} g_{f+20}(z) - \frac{1}{2} g_{f+16}(z) - \frac{1}{2} g_{f+14}(z) + \frac{1}{4} g_{f+12}(z) + g_{f+10}(z) \right. \\
& \quad \left. + \frac{1}{4} g_{f+8}(z) - \frac{1}{2} g_{f+6}(z) - \frac{1}{2} g_{f+4}(z) + \frac{1}{4} g_f(z) \right] \\
& + \omega_2^3 \omega_4 \left[\frac{1}{6} g_{f+20}(z) - \frac{1}{2} g_{f+16}(z) + \frac{1}{3} g_{f+12}(z) + \frac{1}{3} g_{f+18}(z) - \frac{1}{2} g_{f+4}(z) + \frac{1}{6} g_f(z) \right] \\
& + \omega_2^5 \left[\frac{1}{120} g_{f+20}(z) - \frac{1}{24} g_{f+16}(z) + \frac{1}{12} g_{f+12}(z) - \frac{1}{12} g_{f+8}(z) \right. \\
& \quad \left. + \frac{1}{24} g_{f+4}(z) - \frac{1}{120} g_f(z) \right],
\end{aligned}$$

$$\begin{aligned}
S_{11}(z) = & \omega_{11} [g_{f+22}(z) - g_f(z)] + \omega_9 \omega_2 [g_{f+22}(z) - g_{f+18}(z) - g_{f+4}(z) + g_f(z)] \\
& + \omega_8 \omega_3 [g_{f+22}(z) - g_{f+16}(z) - g_{f+6}(z) + g_f(z)] \\
& + \omega_7 \omega_2^2 \left[\frac{1}{2} g_{f+22}(z) - g_{f+18}(z) + \frac{1}{2} g_{f+14}(z) - \frac{1}{2} g_{f+8}(z) + g_{f+4}(z) - \frac{1}{2} g_f(z) \right] \\
& + \omega_7 \omega_4 [g_{f+22}(z) - g_{f+14}(z) - g_{f+8}(z) + g_f(z)] \\
& + \omega_6 \omega_5 [g_{f+22}(z) - g_{f+12}(z) - g_{f+10}(z) + g_f(z)] \\
& + \omega_6 \omega_3 \omega_2 [g_{f+22}(z) - g_{f+18}(z) - g_{f+16}(z) + g_{f+12}(z) - g_{f+10}(z) \\
& \quad + g_{f+6}(z) + g_{f+4}(z) - g_f(z)] \\
& + \omega_5 \omega_4 \omega_2 [g_{f+22}(z) - g_{f+18}(z) - g_{f+14}(z) - g_{f+12}(z) + g_{f+10}(z) + g_{f+8}(z) \\
& \quad + g_{f+4}(z) - g_f(z)] \\
& + \omega_5 \omega_3^2 \left[\frac{1}{2} g_{f+22}(z) - g_{f+16}(z) - \frac{1}{2} g_{f+12}(z) + \frac{1}{2} g_{f+10}(z) \right. \\
& \quad \left. + g_{f+6}(z) - \frac{1}{2} g_f(z) \right] \\
& + \omega_5 \omega_2^3 \left[\frac{1}{6} g_{f+22}(z) - \frac{1}{2} g_{f+18}(z) + \frac{1}{2} g_{f+14}(z) - \frac{1}{6} g_{f+12}(z) \right. \\
& \quad \left. - \frac{1}{6} g_{f+10}(z) + \frac{1}{2} g_{f+8}(z) - \frac{1}{2} g_{f+4}(z) + \frac{1}{6} g_f(z) \right] \\
& + \omega_4^2 \omega_3 \left[\frac{1}{2} g_{f+22}(z) - \frac{1}{2} g_{f+16}(z) - g_{f+14}(z) + g_{f+8}(z) + \frac{1}{2} g_{f+6}(z) - \frac{1}{2} g_f(z) \right] \\
& + \omega_4 \omega_3 \omega_2^2 \left[\frac{1}{2} g_{f+22}(z) - g_{f+18}(z) - \frac{1}{2} g_{f+16}(z) + g_{f+12}(z) + g_{f+10}(z) - \frac{1}{2} g_{f+6}(z) \right. \\
& \quad \left. - g_{f+4}(z) + \frac{1}{2} g_f(z) \right] \\
& + \omega_3^3 \omega_2 \left[\frac{1}{6} g_{f+22}(z) - \frac{1}{6} g_{f+18}(z) - \frac{1}{2} g_{f+16}(z) + \frac{1}{2} g_{f+12}(z) \right.
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2}g_{f+10}(z) - \frac{1}{2}g_{f+6}(z) - \frac{1}{6}g_{f+4}(z) + \frac{1}{6}g_f(z) \\
& + \omega_3\omega_2^4 \left[\frac{1}{24}g_{f+22}(z) - \frac{1}{6}g_{f+18}(z) - \frac{1}{24}g_{f+16}(z) + \frac{1}{4}g_{f+14}(z) \right. \\
& \quad + \frac{1}{6}g_{f+12}(z) - \frac{1}{6}g_{f+10}(z) - \frac{1}{4}g_{f+8}(z) + \frac{1}{24}g_{f+6}(z) \\
& \quad \left. + \frac{1}{6}g_{f+4}(z) - \frac{1}{24}g_f(z) \right],
\end{aligned}$$

$$\begin{aligned}
S_{12}(z) = & \omega_{12}[g_{f+24}(z) - g_f(z)] + \omega_{10}\omega_2[g_{f+24}(z) - g_{f+20}(z) - g_{f+4}(z) + g_f(z)] \\
& + \omega_9\omega_3[g_{f+24}(z) - g_{f+18}(z) - g_{f+6}(z) + g_f(z)] + \omega_8\omega_4[g_{f+24}(z) - g_{f+16}(z) \\
& \quad - g_{f+8}(z) + g_f(z)] \\
& + \omega_8\omega_2^2 \left[\frac{1}{2}g_{f+24}(z) - g_{f+20}(z) + \frac{1}{2}g_{f+16}(z) - \frac{1}{2}g_{f+8}(z) + g_{f+4}(z) - \frac{1}{2}g_f(z) \right] \\
& + \omega_7\omega_5[g_{f+24}(z) - g_{f+14}(z) - g_{f+10}(z) + g_f(z)] \\
& + \omega_7\omega_3\omega_2[g_{f+24}(z) - g_{f+20}(z) - g_{f+18}(z) + g_{f+14}(z) - g_{f+10}(z) + g_{f+6}(z) \\
& \quad + g_{f+4}(z) - g_f(z)] \\
& + \omega_6^2 \left[\frac{1}{2}g_{f+24}(z) - g_{f+12}(z) + \frac{1}{2}g_f(z) \right] \\
& + \omega_6\omega_4\omega_2[g_{f+24}(z) - g_{f+20}(z) - g_{f+16}(z) + g_{f+8}(z) + g_{f+4}(z) - g_f(z)] \\
& + \omega_6\omega_2^3 \left[\frac{1}{2}g_{f+24}(z) - \frac{1}{2}g_{f+20}(z) + \frac{1}{2}g_{f+16}(z) - \frac{1}{3}g_{f+12}(z) + \frac{1}{2}g_{f+8}(z) \right. \\
& \quad \left. - \frac{1}{2}g_{f+4}(z) + \frac{1}{6}g_f(z) \right] \\
& + \omega_6\omega_3^2 \left[\frac{1}{2}g_{f+24}(z) - g_{f+18}(z) + g_{f+6}(z) - \frac{1}{2}g_f(z) \right] \\
& + \omega_5^2\omega_2 \left[\frac{1}{2}g_{f+24}(z) - \frac{1}{2}g_{f+20}(z) - g_{f+14}(z) + g_{f+10}(z) + \frac{1}{2}g_{f+4}(z) - \frac{1}{2}g_f(z) \right] \\
& + \omega_5\omega_4\omega_3[g_{f+24}(z) - g_{f+18}(z) - g_{f+16}(z) - g_{f+14}(z) + g_{f+10}(z) + g_{f+8}(z) \\
& \quad + g_{f+6}(z) - g_f(z)]
\end{aligned}$$

$$\begin{aligned}
& + \omega_5 \omega_3 \omega_2^2 \left[\frac{1}{2} g_{f+24}(z) - g_{f+20}(z) - \frac{1}{2} g_{f+18}(z) + \frac{1}{2} g_{f+12}(z) + \frac{1}{2} g_{f+14}(z) \right. \\
& \quad \left. + \frac{1}{2} g_{f+10}(z) + \frac{1}{2} g_{f+8}(z) - \frac{1}{2} g_{f+6}(z) - g_{f+4}(z) + \frac{1}{2} g_f(z) \right] \\
& + \omega_4^3 \left[\frac{1}{6} g_{f+24}(z) - \frac{1}{2} g_{f+16}(z) + \frac{1}{2} g_{f+8}(z) - \frac{1}{6} g_f(z) \right] \\
& + \omega_4^2 \omega_2^2 \left[\frac{1}{4} g_{f+24}(z) - \frac{1}{2} g_{f+20}(z) - \frac{1}{4} g_{f+16}(z) + g_{f+12}(z) - \frac{1}{4} g_{f+8}(z) \right. \\
& \quad \left. - \frac{1}{2} g_{f+4}(z) + \frac{1}{4} g_f(z) \right] \\
& + \omega_4 \omega_3^2 \omega_2 \left[\frac{1}{2} g_{f+24}(z) - \frac{1}{2} g_{f+20}(z) - g_{f+18}(z) - \frac{1}{2} g_{f+16}(z) + g_{f+14}(z) \right. \\
& \quad \left. + g_{f+12}(z) + g_{f+10}(z) - \frac{1}{2} g_{f+8}(z) - g_{f+6}(z) - \frac{1}{2} g_{f+4}(z) + \frac{1}{2} g_f(z) \right] \\
& + \omega_4 \omega_2^4 \left[\frac{1}{24} g_{f+24}(z) - \frac{1}{6} g_{f+20}(z) + \frac{5}{24} g_{f+16}(z) - \frac{5}{24} g_{f+8}(z) + \frac{1}{6} g_{f+4}(z) \right. \\
& \quad \left. - \frac{1}{24} g_f(z) \right] \\
& + \omega_3^4 \left[\frac{1}{24} g_{f+24}(z) - \frac{1}{6} g_{f+18}(z) + \frac{1}{4} g_{f+12}(z) - \frac{1}{6} g_{f+6}(z) + \frac{1}{24} g_f(z) \right] \\
& + \omega_3^2 \omega_2^3 \left[\frac{1}{12} g_{f+24}(z) - \frac{1}{4} g_{f+20}(z) - \frac{1}{6} g_{f+18}(z) + \frac{1}{4} g_{f+16}(z) \right. \\
& \quad \left. + \frac{1}{2} g_{f+14}(z) - \frac{1}{2} g_{f+10}(z) - \frac{1}{4} g_{f+8}(z) + \frac{1}{6} g_{f+6}(z) + \frac{1}{4} g_{f+4}(z) - \frac{1}{12} g_f(z) \right] \\
& + \omega_2^6 \left[\frac{1}{720} g_{f+24}(z) - \frac{1}{120} g_{f+20}(z) + \frac{1}{48} g_{f+16}(z) - \frac{1}{36} g_{f+12}(z) \right. \\
& \quad \left. + \frac{1}{48} g_{f+8}(z) - \frac{1}{120} g_{f+4}(z) + \frac{1}{720} g_f(z) \right].
\end{aligned}$$